

Dino2-DR: A Trustworthy and Explainable Vision Transformer for Cross-Domain Diabetic Retinopathy Grading

Lucia Cascone Luigi Di Biasi Giuseppe Genito Michele Nappi

University of Salerno

Department of Computer Science

Via Giovanni Paolo II, 132, 84084 Fisciano (SA), Italy

lcascone@unisa.it, ldibiasi@unisa.it, g.genito1@studenti.unisa.it, mnappi@unisa.it

Abstract

Deep learning models for diabetic retinopathy (DR) classification often exhibit performance degradation under domain shift, limiting their reliability in heterogeneous clinical environments. We introduce Dino2-DR, a robust and interpretable framework for DR severity grading. Built upon the self-supervised DINOv2 Vision Transformer, our approach is assessed under two complementary evaluation protocols that reflect realistic deployment conditions. For the Leave-One-Domain-Out protocol, we train six independent models, systematically withholding one fundus dataset (APTOS, DeepDRiD, FGADR, IDRiD, Messidor2, RLDR) for out-of-domain testing. This protocol achieves 85.73% AUC, 62.47% accuracy, and 50.15% F1-score averaged across all target domains, outperforming existing CNN and ViT-based approaches. The Fixed-Source Multi-Target (FSMT) protocol trains a single model on EyePACS + DDR and evaluates generalization to six external domains, yielding 87.83% AUC, 66.07% accuracy, and 50.46% F1-score. Beyond strong predictive performance, Dino2-DR prioritizes explainability and reliability: attention-rollout visualizations spotlight clinically relevant regions, and a light optional temperature-scaling step preserves a low calibration error (ECE less than 0.05) on every domain. An interactive interface allows clinicians to inspect predictions and associated evidence, facilitating potential adoption in routine clinical screening. Code: <https://github.com/CASALab-Unisa/Dino2-DR>

1. Introduction

Diabetic retinopathy (DR) is a common microvascular complication of diabetes and a leading cause of preventable vision impairment among working-age adults. The condition progresses from early non-proliferative stages—marked by microaneurysms and vascular leakage—to more advanced

proliferative forms involving neovascularization and potential retinal detachment. Regular fundus screening is essential for early detection and risk reduction, yet large-scale implementation remains logistically challenging.

Traditional DR screening relies on manual grading by ophthalmologists or trained readers, introducing diagnostic bottlenecks and limiting access, particularly in low-resource settings. Although deep learning models have achieved strong performance under controlled conditions [1], their deployment in real-world scenarios is limited by a key challenge: poor generalization across diverse imaging conditions, acquisition devices, and patient populations—a phenomenon known as *domain shift*.

The generalization challenge. Domain shift leads to a marked drop in performance when models are applied outside their training distribution, undermining their reliability in real-world settings. The variability in the appearance, contrast, and image quality of the lesions [2] - due to differences in the acquisition protocols and devices - poses significant obstacles to automated DR classification. As a result, models trained on a single dataset often fail to retain their performance when evaluated on external data, with noticeable declines in both classification accuracy and discrimination capability [3], [4]. Although this issue is being progressively addressed through recent methodological advances [2], leading to measurable improvements in predictive performance, a fundamental barrier to clinical adoption persists: the intrinsic opacity of deep learning models.

Opacity as a barrier to clinical trust. In high-stakes medical scenarios, predictive accuracy is not sufficient on its own. For AI systems to be meaningfully integrated into clinical workflows, their decisions must be understandable, justifiable, and aligned with established medical knowledge. This is essential to foster trust among clinicians and regulatory bodies, who need to evaluate not only what a model predicts, but also why [5]. Surveys consistently reveal this gap: few ophthalmologists re-

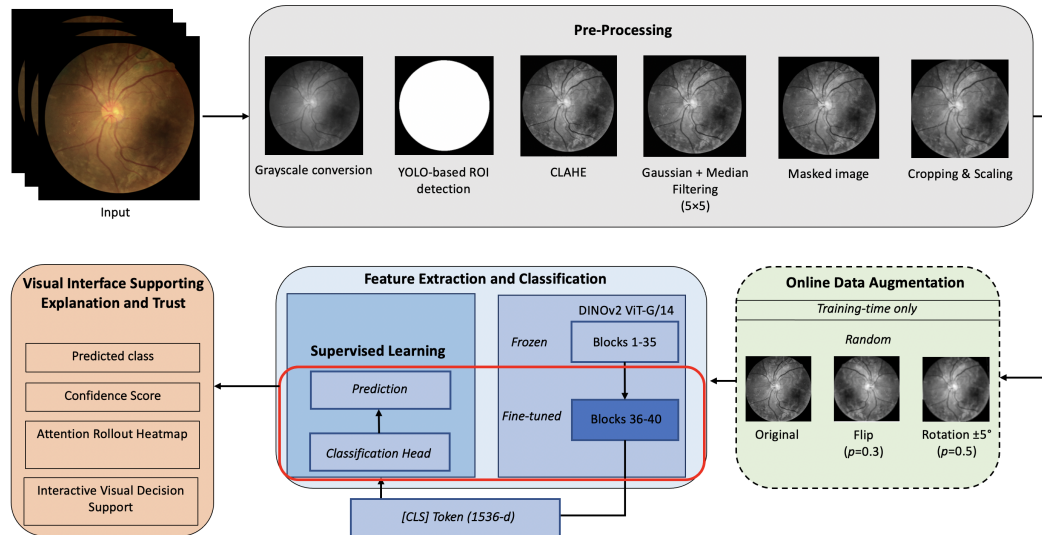


Figure 1. Dino2-DR architecture. The system includes: (1) preprocessing to standardize fundus images across domains; (2) online data augmentation for improved generalization; (3) a fine-tuned DINOv2 Vision Transformer for feature extraction; (4) a classification head for DR severity prediction; and (5) an explainability module using Attention Rollout, integrated into an interactive interface for clinician feedback and trust.

port confidence in fully autonomous DR systems, primarily due to limited explainability and the absence of uncertainty quantification [6]. In this setting, explainability refers to the model’s ability to provide clear and accessible reasons for its outputs, while trust reflects the confidence that users—particularly healthcare professionals—place in the model’s consistency, reliability, and clinical relevance. These aspects are closely related: explainability enables understanding, which is essential for trust; without it, even accurate models may be seen as opaque and unsuitable for clinical adoption.

Contributions. This work aims to address these two fundamental challenges: limited cross-domain generalization and insufficient explainability. As illustrated in Figure 1, we present Dino2-DR, a vision transformer-based framework that integrates cross-domain robustness with explainability, aiming to enhance clinical trust in DR classification.

- We introduce Dino2-DR, a vision-transformer framework that extends DINOv2’s self-supervised representations with lightweight, domain-aware fine-tuning. Tested under a rigorous domain-generalization (DG) protocol spanning six public DR datasets, Dino2-DR achieves consistently higher AUC, accuracy, and F1 than competing baselines when evaluated on unseen data.
- To address explainability, we integrate an attention-based visualization module that makes the model’s decision process more transparent. Specifically, we measure how well the model’s attention maps align with clinically relevant regions identified by experts, providing insight into its focus on meaningful diagnostic cues.

- To support this process, we develop a web-based interface that enables ophthalmologists to draw regions of interest directly on the fundus images and receive real-time feedback. The tool displays both the model’s prediction of DR severity—together with its confidence score—and the overlap between the user-annotated areas and the model’s attention focus. This interactive setup encourages critical review of the model’s output and promotes trust by aligning automated decisions with expert expectations.
- We validate the reliability of the predicted scores by assessing their calibration, reporting an expected calibration error (ECE) below 0.05 across held-out domains.

The remainder of the paper is organized as follows. Section 2 reviews related work and situates our approach within the existing literature. Section 3 details the proposed methodology, and Section 4 describes the experimental setup. Results and analysis are presented in Section 5. Finally, Section 6 concludes the paper with a summary of key findings, limitations, and directions for future work.

2. Related Work

The challenge of developing robust DR grading systems that generalize across diverse clinical environments represents a critical intersection of computer vision and healthcare AI. Domain shifts in medical imaging arise from fundamental differences in acquisition protocols, imaging equipment, patient demographics, and annotation standards across healthcare institutions [7]. These variations create significant performance degradation when models trained

on one clinical dataset are deployed in different healthcare settings, limiting the practical applicability of AI-based diagnostic systems as already discussed in Section 1. Early approaches to the problem were based on convolutional neural networks (CNNs), typically trained under the assumption that training and test data were drawn from the same distribution. Atwany et al. [8] were among the first to question this assumption by introducing a benchmark for evaluating CNNs under domain shift. Che et al. [2] formalized the problem further by proposing *GDRBench*, which unifies eight public DR datasets and introduces the now-standard *leave-one-domain-out* (LODO) evaluation protocol. In this setting, a model is trained on $K-1$ domains and tested on the held-out domain, simulating deployment in unseen clinical settings. GDRNet, their ResNet50-based baseline with domain-aware regularization, achieved improved robustness but remained limited by the local and inductive biases inherent to CNNs.

More recently, Vision Transformers (ViTs) [9] have since emerged as a powerful alternative due to their ability to capture long-range dependencies and global lesion context—key properties for DR grading. MIL-ViT [10] was among the first to apply ViT architectures in this context, adapting them to a multiple-instance learning setting and achieving superior results on *GDRBench* compared to CNN-based models. Building on this foundation, the Grade-Aware De-stylization (GAD) framework [11] paired a Swin Transformer [12] encoder with two modules specifically designed for DG: (i) a grade-aware contrastive learning objective based on global channel similarity, which encourages intra-class compactness across domains, and (ii) a multi-scale de-stylization component that explicitly removes style information through deep whitening transformations. Under the LODO protocol, GAD consistently outperforms MIL-ViT and previous state-of-the-art methods in terms of AUC, accuracy, and F1-score, with minimal increase in computational cost. This positions GAD as the current leading ViT-based solution for robust DR grading with DG protocol.

Vision-language pretraining has also been explored as a means of achieving zero-shot generalization. CLIP-DR [13] reframed DR grading as a text–image alignment task using the CLIP model [14]. By matching fundus images to textual descriptions of DR severity, this approach demonstrated promising zero-shot transfer across domains. However, it remains sensitive to prompt formulation and lacks mechanisms to ensure focus on clinically meaningful features.

In summary, the field has progressed from CNN-based regularization strategies toward transformer-based architectures with explicit mechanisms for domain adaptation and de-stylization. The GAD framework exemplifies this shift, combining ViT capacity with domain-aware training objectives to achieve state-of-the-art performance under the stan-

dard LODO protocol.

In Section 5, our results indicate that the proposed approach extends this line of work and achieves improved performance across most domains and evaluation metrics.

3. Methodology

Figure 1 illustrates the architecture of the Dino2-DR system, highlighting the main processing stages and the integration of the explainability interface.

Pre-Processing. To enable robust cross-domain generalization, we designed a preprocessing pipeline that standardizes fundus images while preserving clinical features. It addresses three major challenges: variations in color and lighting, geometric inconsistencies in field-of-view and resolution, and dataset-specific artifacts that may bias learning.

Initial preprocessing converts all images to grayscale to emphasize structural features and reduce acquisition-related color variability. Due to heterogeneity in input resolutions and aspect ratios, black padding is applied to transform each image into a square format prior to region of interest (ROI) detection using the YOLO model [15].

To isolate the retinal region, we applied Gaussian blurring (kernel size 15×15) to reduce noise, followed by global thresholding (value 30) to highlight high-luminance areas. Morphological opening (elliptical kernel 15×15) and closing (kernel 35×35 , two iterations) were used to refine the mask. The largest connected component was extracted, and its convex hull was computed to define the final ROI. Following ROI extraction, each image was processed to enhance local contrast and reduce noise. CLAHE was applied with fixed parameters to normalize contrast across datasets, followed by 5×5 Gaussian and median filtering to reduce high-frequency and impulsive noise, respectively.

The final preprocessing stage addresses geometric standardization to ensure consistent fundus representation across the dataset. The segmented fundus mask is applied to the contrast-enhanced images, followed by cropping to the mask boundaries to eliminate background regions. To accommodate anatomical size variations while maintaining uniformity, we implement a strategic scaling approach: Each cropped fundus is resized so that its largest dimension occupies 90% of the target pixel dimension. This design choice ensures that the fundus consistently occupies the same relative image area while providing sufficient margin to prevent boundary artifacts and accommodate natural size variations.

Final images are centered on 504×504 pixel canvases and stored as 8-bit grayscale tensors, ready for input into the downstream model, whose patch partitioning is aligned with this resolution.

Online Data Augmentation. To promote generalization and mitigate overfitting without introducing distributional shifts, we employed a conservative online data augmenta-

Table 1. Class distribution of DR severity across datasets.

Dataset	No DR	Mild	Moderate	Severe	Proliferative
APTOS	1,421	247	616	116	183
DDR	6,265	630	4,477	236	913
DeepDRiD	680	356	392	392	180
EyePACS	65,339	6,204	13,153	2,087	1,914
FGADR	101	212	595	647	287
IDRiD	168	25	157	86	58
Messidor2	895	189	256	59	34
RLDR	166	337	929	99	62

tion strategy. Rather than augmenting the dataset offline, stochastic transformations were applied on-the-fly during training. Each input sample was independently subjected to either a horizontal flip with a probability of 0.3 or a random rotation within a range of $\pm 5^\circ$ with probability 0.5. Transformations were applied independently to each sample; in some cases, no augmentation occurred. This minimal augmentation policy was selected following preliminary validation experiments, which indicated that more aggressive transformations did not improve performance and occasionally introduced undesirable artifacts.

DINOv2-DR. We employed DINOv2 [16] in Giant configuration (ViT-G/14) as feature extractor, comprising 1.1 billion parameters pretrained. The model generates 1536-dimensional embeddings from the [CLS], which are processed through a classification head: linear projection (1536→512), ReLU activation, dropout (30%), and final classification layer (512→5 classes). Our transfer learning strategy involved selective fine-tuning of the final five transformer blocks while freezing remaining layers to preserve pretrained representations. The model was trained using AdamW optimizer with differential learning rates (1×10^{-3} for classification head, 1×10^{-6} for unfrozen blocks), batch size 4, and Cross-Entropy loss. Early stopping on validation loss (patience=5) typically converged within 15 epochs.

Trustworthiness. To evaluate the reliability of the model’s probabilistic outputs, we adopt post-hoc *Temperature Scaling* (TS) [17], a parametric calibration method that adjusts the confidence scores without altering class predictions. Let $\mathbf{z} = g^\theta(\mathbf{x}) \in \mathbb{R}^K$ denote the vector of pre-softmax logits for input $\mathbf{x}$ to the trained model g^θ , where K is the number of classes. TS applies a scalar temperature parameter $T > 0$ to obtain the calibrated softmax distribution: $\hat{p}_i = \frac{\exp(z_i/T)}{\sum_{k=1}^K \exp(z_k/T)}$ for $i = 1, \dots, K$. The optimal value of T is obtained by minimizing the Negative Log-Likelihood (NLL) over a held-out validation set drawn from the source domain: $T^* = \arg \min_{T>0} \mathcal{L}_{\text{NLL}}(T) = -\sum_{j=1}^n \log \hat{p}_{y_j}^{(j)}$, where $\hat{p}_{y_j}^{(j)}$ is the calibrated confidence assigned to the true label y_j for sample j .

To assess calibration, we report both NLL and the Expected Calibration Error (ECE), defined as: $\text{ECE} = \sum_{m=1}^M \frac{|B_m|}{n} |\text{acc}(B_m) - \text{conf}(B_m)|$, where B_m denotes the set of samples whose predicted confidence falls into the

m -th bin, n is the total number of samples, and $M = 15$ is the number of equal-width bins used. Here, $\text{acc}(B_m)$ is the empirical accuracy and $\text{conf}(B_m)$ is the average predicted confidence within the bin.

Explainability. Attention Rollout. To assess whether the model focuses on clinically meaningful features, we devised an explainability pipeline based on Attention Rollout [18], a method tailored for Vision Transformers. For an input image discretised into an $H \times W$ patch grid, recursive head-averaged aggregation yields a saliency map $\mathcal{A} \in [0, 1]^{H \times W}$, where $\mathcal{A}(i, j)$ is the contribution of patch (i, j) to the [CLS] decision. These maps are upsampled and overlaid onto the original image, providing an interpretable visual representation of the model’s focus and supporting visual inspection by clinical users.

Diagnostic Support Interface for Clinicians. To enhance model explainability within a practical clinical setting, we designed a web-based interface that embeds visual explanation into the diagnostic workflow. The system allows ophthalmologists to examine the model’s predictions by overlaying attention-based saliency maps onto retinal fundus images, allowing transparent inspection of the model’s decision rationale. Upon uploading a retinal fundus image, the interface automatically displays the predicted DR class, the associated prediction confidence score, and a saliency heatmap derived from Attention Rollout, overlaid directly on the original image. Clinicians can interactively annotate regions of interest that they consider diagnostically relevant. For each user-defined ROI $R \subseteq \{1, \dots, H\} \times \{1, \dots, W\}$, the system computes a *coverage score* defined as the proportion of pixels within R that exhibit high model attention. Formally, given the attention rollout map $\mathcal{A} \in [0, 1]^{H \times W}$ and a threshold $\tau \in (0, 1)$ derived from a colormap rule (e.g., highlighting red-toned regions), the binary attention mask is defined as $\mathcal{A}_\tau(i, j) = \mathbf{1}(\mathcal{A}(i, j) \geq \tau)$. The coverage for ROI R is then computed as: $\text{Coverage}(R) = \frac{1}{|R|} \sum_{(i,j) \in R} \mathcal{A}_\tau(i, j)$. When a clinician draws a region of interest, the system computes its attention overlap and displays the corresponding value in the top status bar. At the same time, the drawn region is outlined in green if the overlap exceeds a clinically meaningful threshold, or in red otherwise—providing both quantitative feedback and intuitive visual validation. This design enhances explainability by enabling clinicians to actively inspect and quantify the degree to which the model’s internal attention patterns align with human diagnostic expectations. Although the attention heatmap derived from rollout is not a causal explanation of the decision process, it serves as a meaningful proxy to reveal which regions the model focuses on when making predictions. By visualizing and quantifying this focus, the interface fosters transparent reasoning and supports the validation of model behavior against clinical intuition.

Table 2. Comparison of model performance using the LODO evaluation protocol. Metrics include Area Under the Curve (AUC), Accuracy (ACC), and F1-score, expressed as percentages (%). The highest value for each metric and target domain is highlighted in bold. CNN-based baselines are reported as in [2], while ViT-based models follow the implementation details of [11].

Target Metrics	APTOS			DeepDR			FGADR			IDRID			Messidor			RLDR			Average		
	AUC	ACC	F1	AUC	ACC	F1	AUC	ACC	F1	AUC	ACC	F1	AUC	ACC	F1	AUC	ACC	F1	AUC	ACC	F1
<i>CNN based:</i>																					
DRGen [19]	79.9	58.1	40.2	83	38.7	34.1	69.4	41.7	24.7	84.7	44.6	37.4	79	60.1	40.5	79.5	43.1	37.0	79.3	47.7	37.3
Mixup [20]	75.3	62.6	43.2	75.3	29	25.2	66.7	42.3	32.3	78.8	39	27.6	76.7	54.7	32.6	76.9	43.6	37.7	75	45.2	33.1
MixStyle [21]	79	65.8	39.9	76.9	32.9	27.9	71.2	35.8	22.7	83	51.4	39.2	75.2	62.2	36.5	75.5	41.1	31.4	76.8	48.2	32.9
GREEN [22]	75.1	53.8	38.9	76.4	28.1	24.9	65.5	41.3	31.5	79.9	41.3	32.2	75.8	52	36.8	74.8	34	34.4	75.3	41.8	33.1
CABNet [23]	75.8	55.5	39.4	75.2	42.7	31.8	73.2	43.7	34.8	79.2	44.8	37.3	74.2	56.1	34.1	75.8	37	35.6	75.6	46.6	35.5
DDAIG [24]	78	67.1	41	75.6	37.6	32.2	73.6	42	33.8	82.1	37.4	27	76.6	58.4	35.3	75.6	36.1	27.7	76.9	46.4	32.8
ATS [25]	77.1	56.9	38.3	79.4	36.1	31.6	74.7	46.7	33.4	83	41.5	34.9	77.2	64.7	35.8	76.5	37.4	34.9	78	47.2	34.8
Fishr [26]	79.2	66.6	43.4	81.1	48.1	34.4	73.3	44.4	34.4	82.7	40.3	27.6	76.4	65.1	41.1	77.4	36.8	34.7	78.4	50.2	35.9
MDLT [27]	77.3	57.2	41.5	80	39.5	36.2	74.1	45.7	29	81.5	44.2	35.4	75.4	58.9	36.9	75.7	37.6	35	77.3	47.2	35.7
GDRNet [2]	79.9	66.8	46	84.7	53.1	45.3	80.8	45.3	39.4	84	40.3	35.9	83.2	63.4	50.9	82.9	45.8	43.5	82.6	52.5	43.5
<i>Vision-language:</i>																					
CLIP-DR [13]	83.3	-	46.3	89.9	-	45.8	80.7	-	48	84.5	-	41.9	79	-	47.3	78.9	-	41	82.72	-	45.5
<i>ViT based:</i>																					
MIL-ViT [10]	80.4	68.1	46.2	83.8	52.1	41.1	78.4	44.3	31.5	85.3	40.6	39.9	81.9	60.3	45.2	81.5	45.0	41.8	81.9	51.7	41.0
GAD [11]	81.7	70.6	47.0	85.9	62.0	46.9	79.0	50.0	42.1	85.5	50.6	42.3	84.0	63.8	47.3	83.6	47.4	42.2	83.3	57.4	44.6
Dino2-DR	85.60	74.84	52.21	82.96	54.85	46.02	83.40	55.70	48.99	88.33	57.69	52.63	89.24	74.32	49.17	84.85	57.44	51.88	85.73	62.47	50.15

Table 3. Comparison of *Dino2-DR* performance under the two evaluation protocols. In the FSMT setting, the model is trained on a fixed source (EyePACS + DDR) and evaluated on six target domains. In the LODO setting, the model is trained on five domains and evaluated on the held-out sixth. Bold values indicate the best performance per column and metric.

Protocol	APTOS			DeepDRiD			FGADR			IDRID			Messidor2			RLDR			Average		
	AUC	ACC	F1	AUC	ACC	F1	AUC	ACC	F1	AUC	ACC	F1	AUC	ACC	F1	AUC	ACC	F1	AUC	ACC	F1
LODO	85.60	74.84	52.21	82.96	54.85	46.02	83.40	55.70	48.99	88.33	57.69	52.63	89.24	74.32	49.17	84.85	57.44	51.88	85.73	62.47	50.15
FSMT	85.18	79.13	53.39	85.58	55.10	41.54	85.17	52.50	45.15	89.74	65.99	54.76	91.96	79.41	56.42	89.32	64.28	51.47	87.83	66.07	50.46

4. Experimental Setup

Datasets. In this study, we utilize multiple datasets annotated according to the *International Clinical Diabetic Retinopathy Disease Severity Scale* [28], which defines five progressive stages: no apparent retinopathy, mild, moderate, and severe non-proliferative DR, and proliferative DR. These datasets differ substantially in the number of images (Table 1), image quality, resolution, camera type (which significantly impacts Field of View (FOV) and brightness), annotation consistency, and class distribution, all factors that can significantly influence model performance and generalizability. Here the DR datasets involved in the research:

APTOS [29] was released by the Asia Pacific Tele-Ophthalmology Society. Image resolutions vary across subsets (1050×1050 px for training, 640×480 px for testing), and no information is available regarding the imaging devices or FOV. Only the training set is labeled according to the five-stage DR severity scale. After preprocessing—which involved removing corrupted or unmatched samples—a cleaned subset of 2,584 annotated images was retained for analysis.

DDR [30] comprises 12,552 images (702×717 to 5184×3456 px), labeled according to the international DR grading scale. Images were acquired using various devices with a 45° FOV. An additional class for ungradable images is also included.

DeepDRiD [31] includes 2,000 images (1592×1728 to 1976×1984 px), provided for the ISBI 2020 challenge. La-

bels are available for all subsets. No missing or corrupted images were reported.

EyePACS [32] was introduced for the 2015 Kaggle DR challenge. It contains 88,702 color fundus images in JPEG format, with highly imbalanced class distribution and variable resolution (433×298 to 5184×3426 px). Camera model and FOV are not specified.

FGADR [33] is a private dataset of 1,842 fundus images (1280×1280 px) collected by IIAI. All labels are expert-verified; no predefined data splits are provided.

IDRID [34] includes 516 high-resolution images (4288×2848 px), captured with a 50° FOV using a Kowa VX-10 camera. Labels were provided by expert ophthalmologists.

Messidor2 [35, 36] contains 1,744 images (1440×960 to 2304×1536 px) acquired with a Topcon TRC NW6 camera (45° FOV). DR grades are not officially provided but are available via expert annotations on Kaggle. After preprocessing, 1,433 images were retained.

RLDR [37] is a curated subset of EyePACS with 1,593 PNG images (896×896 px). Expert re-annotation improved label quality and lesion-level annotations are available for a subset.

Experimental Protocols. To evaluate the generalization capability of our models across unseen domains, we adopt two complementary experimental protocols: (i) a *LODO* strategy following the DG framework proposed by Gulrajani et al. [38], and (ii) a custom *Fixed-Source, Multi-Target* (FSMT) evaluation protocol introduced in this study.

LODO. Let $\mathcal{D} = \{\mathcal{D}_1, \dots, \mathcal{D}_N\}$ represent a set of N domains, each defined by a distinct distribution over the input space $\mathcal{X}$. From each domain $\mathcal{D}_n$, we sample labeled examples $\{(x_n^k, y_n^k)\}_{k=1}^K$, where $x_n^k \in \mathcal{X}$ and $y_n^k \in \mathcal{Y}$. At each iteration, one domain $\mathcal{D}_t$ is held out as the target test set, and the model is trained on the remaining domains $\mathcal{D}_{\text{train}} = \mathcal{D} \setminus \{\mathcal{D}_t\}$. We randomly reserve 20% of the training data for validation, and use the remaining 80% to train the model. Evaluation is then performed on the held-out domain $\mathcal{D}_t$, which is not observed during training or validation.

FSMT. In this alternative protocol, the training domains are fixed in advance. Specifically, the model is trained solely on the two largest and most diverse datasets (EyePACS and DDR) with 20% of their combined data set aside for validation. The trained model is subsequently evaluated on six unseen datasets: APTOS, IDRiD, Messidor2, DeepDRiD, FGADR, and RLDR. FSMT simulates a realistic deployment scenario in which models must generalize to entirely unseen clinical domains starting from a fixed, representative source.

Reproducibility. All experiments used the official DINOv2 implementation and were run on a single NVIDIA A30 GPU. To enhance reproducibility, we seeded all sources of randomness and enabled deterministic mode in CuDNN. Code is available at: <https://github.com/CASALab-Unisa/Dino2-DR>

5. Results and Discussion

LODO Performance Comparison. Table 2 summarizes the performance of all competing methods under the LODO evaluation protocol. Across all held-out target domains and metrics (AUC, Accuracy, and F1-score), our proposed model *Dino2-DR* achieves the highest average performance: 85.73% AUC, 62.47% Accuracy, and 50.15% F1-score.

Notably, *Dino2-DR* consistently outperforms all CNN-based methods—including GDRNet, the previous state-of-the-art—as well as ViT-based architectures such as MIL-ViT and GAD, particularly in terms of F1-score and accuracy. Improvements are particularly evident in challenging domains such as FGADR and IDRiD, where *Dino2-DR* surpasses GAD by +6.89 and +10.33 points in F1-score, respectively. Similarly, the method achieves the highest F1-score in five out of six held-out domains, confirming its robustness to cross-domain distribution shifts. DeepDR is the only case where *Dino2-DR* does not achieve the best F1-score, trailing GAD by just 0.88 points (46.02 vs. 46.9), thus maintaining competitive performance. However, the model does not achieve the highest AUC on DeepDR, where GAD slightly outperforms it (85.9% vs. 82.96%). Nonetheless, *Dino2-DR* yields a comparable F1-score (46.02 vs. 46.9), indicating a more balanced precision-recall trade-off.

This suggests that while GAD may offer marginally sharper ranking performance, *Dino2-DR* retains better clinical robustness through its overall decision balance. Moreover, compared to MIL-ViT, *Dino2-DR* yields consistent gains across all domains in F1-score and Accuracy, and always matches or exceeds it in AUC. Finally, although CLIP-DR achieves a relatively high average AUC (82.72%), it does not report accuracy values and its average F1-score (45.5%) remains significantly lower than that of *Dino2-DR*, confirming the benefit of domain-aware fine-tuning and explicit supervision offered by *Dino2-DR* in structured multi-domain settings. The confusion-matrix analysis (see Fig. 2) confirms this robustness: the model is highly precise for non-DR cases (grade 0) while maintaining good recall for moderate-to-severe grades (2–4). Most errors occur between adjacent severity levels, mirroring known inter-grader variability and suggesting semantically plausible misclassifications rather than systematic failures. Overall, our experiments show that robust representations, paired with selective fine-tuning, help mitigate domain shift in our benchmarks under the LODO protocol, without the need for complex auxiliary modules or handcrafted adaptations.

Performance Comparison between LODO and FSMT Protocols. Table 3 presents a comparative analysis of the two evaluation protocols adopted in our study. The FSMT setting demonstrates superior average performance across all three metrics: AUC (87.83% vs. 85.73%), Accuracy (66.07% vs. 62.47%), and F1-score (50.46% vs. 50.15%). The AUC and Accuracy improvements—+2.1 and +3.6 respectively—suggest that training on a larger source may enable stronger generalization, likely due to broader data coverage and a more stable optimization landscape. Nonetheless, the LODO configuration remains highly competitive, particularly in terms of F1-score, where the gap is minimal (−0.31). This slight advantage can be attributed to the richer diversity of source domains used in each LODO split, which appears to compensate for the comparatively smaller number of training images. Exposure to a wider range of domain-specific variations may foster more robust decision boundaries, particularly useful in handling rare or borderline cases. These findings highlight a trade-off between training set size and domain heterogeneity: while FSMT benefits from scale and source consistency, LODO may better support generalization under distribution shift by virtue of its domain diversity.

Quantitative assessment of explainability: lesion alignment. Although the attention heatmaps produced by the rollout mechanism are not causal explanations of the model’s decision process, they serve as meaningful proxies to visualize which regions are most influential in the classification. To quantitatively assess the alignment between attention and expert knowledge, we leverage the RLDR dataset [37], which includes lesion-level annotations. Given

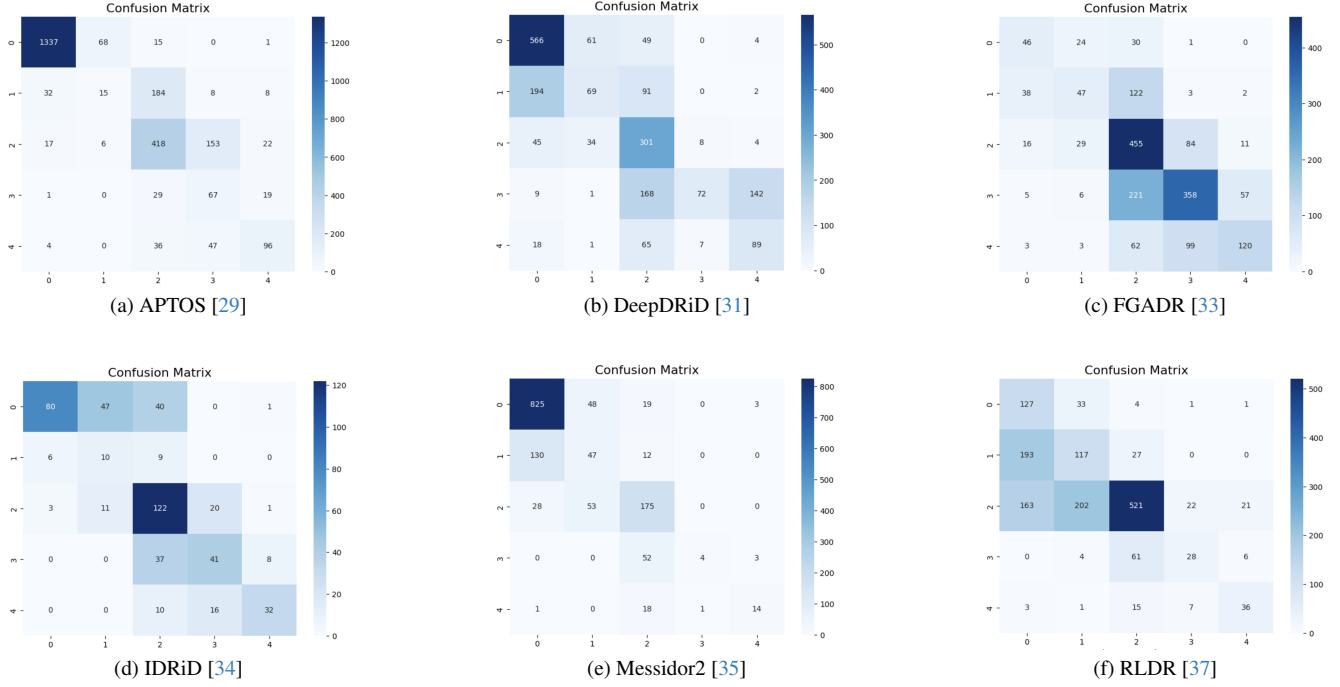


Figure 2. Confusion matrices for each held-out target dataset under the LODO protocol.

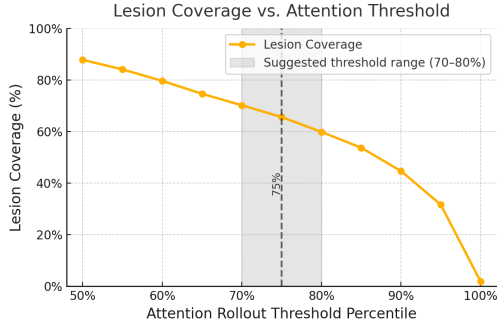


Figure 3. Coverage curve $C(p)$ computed on the RLDR dataset [37], showing the proportion of annotated lesions intersecting with the top- $p\%$ of the attention rollout map.

a percentile threshold $p \in [0, 100]$, we binarised the attention rollout map $\mathcal{A}$ to obtain $\mathcal{A}_p(i, j) = \mathbf{1}[\mathcal{A}(i, j) \geq T_p]$, where T_p denotes the p -th percentile of attention scores. We then computed the lesion-level attention coverage score: $C(p) = \frac{1}{n} \sum_{k=1}^n \mathbf{1}[\text{IoU}(B_k, \mathcal{A}_p) > \varepsilon]$, where $\mathcal{L} = \{B_1, \dots, B_n\}$ are the expert-annotated lesion boxes from the RLDR dataset and $\varepsilon = 0$ is the minimal overlap threshold. As shown in Fig. 3, lower threshold percentiles (i.e., denser attention maps) naturally lead to higher lesion recall. However, we observe that the model often attends beyond annotated lesions, covering adjacent anatomical landmarks such as vessel bifurcations and optic disc margins. This be-

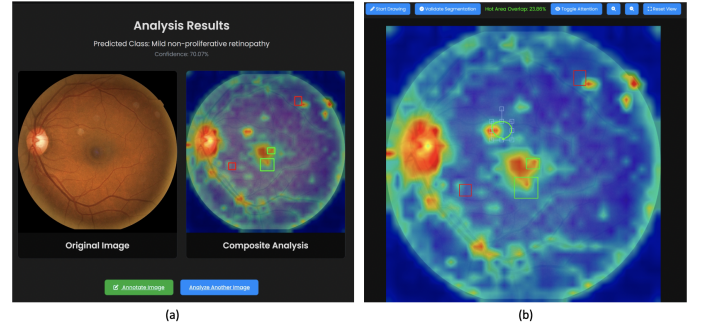


Figure 4. (a) The interface shows the original fundus image alongside an attention-based overlay, highlighting regions deemed relevant by the model. The predicted DR class and confidence score are displayed, and clinicians can annotate regions for feedback. (b) Zoomed-in view of the attention heatmap with clinician-drawn regions (colored boxes), showing the overlap percentage (“Hot Area Overlap”) between user annotations and model attention.

havior suggests that the model does not rely exclusively on explicit pathological features but also integrates clinically relevant contextual cues—mirroring human diagnostic reasoning. This design enhances interpretability by allowing clinicians to actively inspect and quantify how much the model’s attention aligns with their expectations. The interface depicted in Fig. 4 supports this interaction by providing real-time feedback on the overlap between user-defined regions of interest and the attention map, thereby fostering

Table 4. Temperature scaling applied to assess the trustworthiness of model confidence. For each target dataset under the DG protocol, we report the NLL and ECE metrics before and after scaling.

Target Dataset	Before Scaling		Optimal Temp. T	After Scaling	
	NLL ↓	ECE ↓		NLL ↓	ECE ↓
APTOS	0.371	0.022	1.261	0.375	0.040
DeepDRiD	0.365	0.029	1.090	0.370	0.036
FGADR	0.540	0.031	1.245	0.545	0.044
IDRiD	0.458	0.023	1.166	0.465	0.044
Messidor 2	0.616	0.036	1.058	0.620	0.046
RLDR	0.526	0.036	1.096	0.532	0.053

transparency and informed trust in the model’s predictions.

Trustworthiness of confidence estimates. Table 4 reports the results of post-hoc Temperature Scaling applied to assess the reliability of model confidence under the DG protocol. For each target dataset, we report the NLL and ECE before and after scaling, with the optimal temperature T estimated on the source-domain validation set. In all cases, the optimal temperature exceeds 1, reflecting the model’s tendency to produce overconfident predictions—a common phenomenon in large-scale deep networks trained under empirical risk minimization [39]. However, both NLL and ECE values are already remarkably low before calibration ($ECE < 0.05$ across datasets), indicating strong intrinsic reliability of the model’s probabilistic estimates even under domain shift. Interestingly, temperature scaling led to a consistent but negligible increase in both NLL and ECE, suggesting that the model’s confidence estimates are already well-calibrated. This uniform trend suggests that the model’s native confidence estimates are already well-aligned with empirical correctness, and that additional calibration provides limited benefit in this DG setting. These findings reinforce the trustworthiness of our model’s confidence outputs and highlight the robustness of the training strategy adopted.

Ablation Study. To quantify the effect of task-specific fine-tuning, we compared two configurations of DINOv2 [16]: (i) a frozen version used solely as a feature extractor, and (ii) a selectively fine-tuned version where the last five transformer blocks were updated during training. Both models were trained using the same source domains (APTOS, RLDR, Messidor2, IDRiD, and DeepDRiD), and evaluated on the unseen target domain FGADR, following the DG protocol. This comparison isolates the gain provided by aligning pretrained representations to the DR classification task. As shown in Table 5, fine-tuning significantly improves performance across all metrics. In particular, the F1-score nearly doubles, indicating that task-specific adaptation is crucial for handling inter-domain variability and class imbalance in real-world diabetic retinopathy datasets.

Table 5. Comparison of DINOv2 performance with and without fine-tuning on the FGADR test set.

Configuration	AUC	Accuracy	F1-score
Frozen (no fine-tuning)	78.37	34.80	19.00
Fine-tuned (last 5 blocks)	81.74	49.24	37.62

6. Conclusions

In this work, we introduced Dino2-DR, a robust and interpretable vision transformer framework for cross-domain DR severity classification. Leveraging DINOv2’s self-supervised representations and lightweight task-specific fine-tuning, the model consistently outperformed previous CNN- and ViT-based baselines—including GAD and MIL-ViT—under two rigorous evaluation protocols (LODO and FSMT), achieving state-of-the-art performance across six diverse public datasets. Beyond accuracy, we focused on trustworthiness and explainability by providing calibrated confidence scores and attention-based visualizations. An interactive interface enhances clinician engagement by aligning model outputs with human reasoning, fostering transparency in decision-making.

Limitations. Nonetheless, despite surpassing even recent state-of-the-art benchmarks, the current performance levels—particularly in F1-score and error patterns around adjacent severity grades—are not yet sufficient for safe and autonomous clinical deployment. Challenges such as inter-grader variability, limited representation of rare conditions, and residual domain shift effects remain open issues.

Future works. In future work, we plan to further improve generalization by expanding training diversity and sample coverage, as suggested by our comparative analysis of domain scale (FSMT) versus variability (LODO). In addition, we envision the integration of a human-in-the-loop framework during training, where expert feedback—collected through the interactive interface—can guide fine-tuning and correction of the model’s attention patterns. This hybrid strategy not only enhances explainability but also transforms the interface into an active learning channel, ultimately aiming to bridge the gap between algorithmic performance and clinical usability.

Acknowledgement

This research has been supported by Project PNC 0000001 D3-4 Health - CUP B53C22006090001 through National Plan for Complementary Investments to the NRRP, Funded by the European Union - NextGenerationEU.

References

- [1] Santoshkumar Ainapur and Virupakshappa Patil. Automated diabetic retinopathy detection using a multi-step framework

- with stacked ensemble-based classification model. *Expert Systems with Applications*, page 127709, 2025. 1
- [2] Haoxuan Che, Yuhan Cheng, Haibo Jin, and Hao Chen. Towards generalizable diabetic retinopathy grading in unseen domains. In *International Conference on Medical Image Computing and Computer-Assisted Intervention*, pages 430–440. Springer, 2023. 1, 3, 5
 - [3] Varun Gulshan, Lily Peng, Marc Coram, Martin C Stumpe, Derek Wu, Arunachalam Narayanaswamy, Subhashini Venugopalan, Kasumi Widner, Tom Madams, Jorge Cuadros, et al. Development and validation of a deep learning algorithm for detection of diabetic retinopathy in retinal fundus photographs. *jama*, 316(22):2402–2410, 2016. 1
 - [4] Sharon Chokuwa and Muhammad H Khan. Generalizing across domains in diabetic retinopathy via variational autoencoders. In *International Conference on Medical Image Computing and Computer-Assisted Intervention*, pages 265–274. Springer, 2023. 1
 - [5] Valentina Sarao, Daniele Veritti, Axel De Nardin, Micaela Misciagna, Gianluca Foresti, and Paolo Lanzetta. Explainable artificial intelligence model for the detection of geographic atrophy using colour retinal photographs. *BMJ Open Ophthalmology*, 8(1), 2023. 1
 - [6] Hesam Hashemian, Tunde Peto, Renato Ambrósio Jr, Imre Lengyel, Rahele Kafieh, Ahmed Muhammed Noori, and Masoud Khorrami-Nejad. Application of artificial intelligence in ophthalmology: An updated comprehensive review. *Journal of Ophthalmic & Vision Research*, 19(3):354, 2024. 2
 - [7] Kaiyang Zhou, Ziwei Liu, Yu Qiao, Tao Xiang, and Chen Change Loy. Domain generalization: A survey. *IEEE transactions on pattern analysis and machine intelligence*, 45(4):4396–4415, 2022. 2
 - [8] Mohammad Atwany and Mohammad Yaqub. Drgen: domain generalization in diabetic retinopathy classification. In *International Conference on Medical Image Computing and Computer-Assisted Intervention*, pages 635–644. Springer, 2022. 3
 - [9] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, G Heigold, S Gelly, et al. An image is worth 16x16 words: Transformers for image recognition at scale. In *International Conference on Learning Representations*, 2020. 3
 - [10] Qi Bi, Xu Sun, Shuang Yu, Kai Ma, Cheng Bian, Munan Ning, Nanjun He, Yawen Huang, Yuexiang Li, Hanruo Liu, et al. Mil-vit: A multiple instance vision transformer for fundus image classification. *Journal of Visual Communication and Image Representation*, 97:103956, 2023. 3, 5
 - [11] Qi Bi, Jingjun Yi, Hao Zheng, Haolan Zhan, Yawen Huang, Wei Ji, Yuexiang Li, and Yefeng Zheng. Gad: Domain generalized diabetic retinopathy grading by grade-aware de-stylization. *Pattern Recognition*, 164:111484, 2025. 3, 5
 - [12] Ze Liu, Yutong Lin, Yue Cao, Han Hu, Yixuan Wei, Zheng Zhang, Stephen Lin, and Baining Guo. Swin transformer: Hierarchical vision transformer using shifted windows. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 10012–10022, 2021. 3
 - [13] Qinkai Yu, Jianyang Xie, Anh Nguyen, He Zhao, Jiong Zhang, Huazhu Fu, Yitian Zhao, Yalin Zheng, and Yanda Meng. Clip-dr: Textual knowledge-guided diabetic retinopathy grading with ranking-aware prompting. In *International Conference on Medical Image Computing and Computer-Assisted Intervention*, pages 667–677. Springer, 2024. 3, 5
 - [14] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, et al. Learning transferable visual models from natural language supervision. In *International conference on machine learning*, pages 8748–8763. PmlR, 2021. 3
 - [15] Hamid R Alsanad, Amin Z Sadik, Osman N Ucan, Muhammad Ilyas, and Oguz Bayat. Yolo-v3 based real-time drone detection algorithm. *Multimedia tools and applications*, 81(18):26185–26198, 2022. 3
 - [16] Maxime Oquab, Timothée Darcet, Théo Moutakanni, Huy Vo, Marc Szafraniec, Vasil Khalidov, Pierre Fernandez, Daniel Haziza, Francisco Massa, Alaaeldin El-Nouby, et al. Dinov2: Learning robust visual features without supervision. *Transactions on Machine Learning Research Journal*, pages 1–31, 2024. 4, 8
 - [17] Deng-Bao Wang, Lei Feng, and Min-Ling Zhang. Rethinking calibration of deep neural networks: Do not be afraid of overconfidence. *Advances in Neural Information Processing Systems*, 34:11809–11820, 2021. 4
 - [18] Samira Abnar and Willem Zuidema. Quantifying attention flow in transformers. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 4190–4197, 2020. 4
 - [19] Mohammad Atwany and Mohammad Yaqub. Drgen: Domain generalization in diabetic retinopathy classification. In Linwei Wang, Qi Dou, P. Thomas Fletcher, Stefanie Speidel, and Shuo Li, editors, *Medical Image Computing and Computer Assisted Intervention – MICCAI 2022*, 2022. 5
 - [20] Hongyi Zhang, Moustapha Cisse, Yann N Dauphin, and David Lopez-Paz. mixup: Beyond empirical risk minimization. In *International Conference on Learning Representations*, 2018. 5
 - [21] Kaiyang Zhou, Yongxin Yang, Yu Qiao, and Tao Xiang. Domain generalization with mixstyle. In *International Conference on Learning Representations*. 5
 - [22] Shaoteng Liu, Lijun Gong, Kai Ma, and Yefeng Zheng. Green: a graph residual re-ranking network for grading diabetic retinopathy. In *International Conference on Medical Image Computing and Computer-Assisted Intervention*, pages 585–594, 2020. 5
 - [23] Along He, Tao Li, Ning Li, Kai Wang, and Huazhu Fu. Cabnet: Category attention block for imbalanced diabetic retinopathy grading. *IEEE Transactions on Medical Imaging*, 40(1):143–153, 2021. 5
 - [24] Kaiyang Zhou, Yongxin Yang, Timothy Hospedales, and Tao Xiang. Deep domain-adversarial image generation for domain generalisation. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 34, pages 13025–13032. Association for the Advancement of Artificial Intelligence (AAAI), 2020. 5

- [25] Zhong Meng, Jinyu Li, Yifan Gong, and Biing-Hwang Juang. Adversarial teacher-student learning for unsupervised domain adaptation. In *2018 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, page 5949–5953. IEEE, April 2018. 5
- [26] Alexandre Rame, Corentin Dancette, and Matthieu Cord. Fishr: Invariant gradient variances for out-of-distribution generalization. In *International Conference on Machine Learning*, pages 18347–18377. PMLR, 2022. 5
- [27] Yuzhe Yang, Hao Wang, and Dina Katabi. On multi-domain long-tailed recognition, imbalanced domain generalization and beyond. In *European conference on computer vision*, pages 57–75. Springer, 2022. 5
- [28] Wilkinson CP, Ferris FL 3rd, Klein RE, and et al. Proposed international clinical diabetic retinopathy and diabetic macular edema disease severity scales. 5
- [29] Asia Pacific Tele-Ophthalmology Society (APTOS). Aptos 2019 blindness detection, 2019. <https://www.kaggle.com/c/aptos2019-blindness-detection/data>. 5, 7
- [30] Tao Li, Yingqi Gao, Kai Wang, Song Guo, Hanruo Liu, and Hong Kang. Diagnostic assessment of deep learning algorithms for diabetic retinopathy screening. *Information Sciences*, 501:511 – 522, 2019. 5
- [31] Ruhan Liu, Xiangning Wang, Qiang Wu, Ling Dai, Xi Fang, Tao Yan, Jaemin Son, Shiqi Tang, Jiang Li, Zijian Gao, Adrian Galdran, J.M. Poorneshwaran, Hao Liu, Jie Wang, Yerui Chen, Prasanna Porwal, Gavin Siew Wei Tan, Xiaokang Yang, Chao Dai, Haitao Song, Mingang Chen, Huating Li, Weiping Jia, Dinggang Shen, Bin Sheng, and Ping Zhang. Deepdrid: Diabetic retinopathy—grading and image quality estimation challenge. *Patterns*, page 100512, 2022. 5, 7
- [32] Will Cukierski. Diabetic retinopathy detection. 5
- [33] Y. Zhou, B. Wang, L. Huang, S. Cui, and L. Shao. A benchmark for studying diabetic retinopathy: Segmentation, grading, and transferability. *IEEE Transactions on Medical Imaging*, 40(3):818–828, 2021. 5, 7
- [34] Prasanna Porwal, Samiksha Pachade, Ravi Kamble, Manesh Kokare, Girish Deshmukh, Vivek Sahasrabuddhe, and Fabrice Meriaudeau. Indian diabetic retinopathy image dataset (idrid), 2018. 5, 7
- [35] Etienne Decencière, Xinchun Zhang, Guy Cazuguel, Bérengère Lay, Béatrice Cochener, Corinne Trone, Philippe Gain, Romain Ordonez, Pascale Massin, Ali Erginay, Brice Charton, and Jean-Charles Klein. Feedback on a publicly distributed database: the Messidor database. *Image Analysis & Stereology*, 33(3):231–234, aug 2014. 5, 7
- [36] Michael D. Abramoff, James C. Folk, Dennis P. Han, J. D. Walker, D. F. Williams, Stephen R. Russell, Pascale Massin, Béatrice Cochener, Philippe Gain, L. Tang, Mathieu Lamard, D. C. Moga, Gwenolé Quéllec, and Meindert Niemeijer. Automated analysis of retinal images for detection of referable diabetic retinopathy. *JAMA Ophthalmology*, 131(3):351–357, mar 2013. 5
- [37] Qijie Wei, Xirong Li, Weihong Yu, Xiao Zhang, Yongpeng Zhang, Bojie Hu, Bin Mo, Di Gong, Ning Chen, Dayong Ding, et al. Learn to segment retinal lesions and beyond. In *2020 25th International Conference on Pattern Recognition (ICPR)*, pages 7403–7410. IEEE, 2021. 5, 6, 7
- [38] Ishaan Gulrajani and David Lopez-Paz. In search of lost domain generalization. In *International Conference on Learning Representations*. 5
- [39] Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q Weinberger. On calibration of modern neural networks. In *International conference on machine learning*, pages 1321–1330. PMLR, 2017. 8